

The Operating System as a Resource Manager

In the **resource manager** view, the operating system is responsible for managing all parts of a computer system and allocating them in an orderly, fair, and controlled way among different programs and users.

Basic Idea

This is a **bottom-up view** of the operating system. Instead of focusing mainly on abstractions, this view focuses on **management of resources**.

Modern computers contain many resources, such as:

- processors,
- main memory,
- timers,
- disks,
- printers,
- network interfaces,
- mice and other I/O devices.

The operating system acts as a **resource manager** by deciding which program gets which resource, when it gets it, and for how long.

Why Resource Management is Necessary

Modern operating systems allow multiple programs to stay in memory and run at the same time. If all of them tried to use the same device directly, the result would be disorder.

For example, if several programs tried to print at the same time on one printer:

- lines from one program could mix with another,
- the output would become confused,
- the printer would not behave in an orderly way.

The operating system prevents this chaos by controlling access to the printer. It may store print output temporarily and send jobs one by one in the correct order.

Without the operating system, shared resources such as the CPU, memory, and printer could easily lead to conflict and confusion.

Main Duties of the OS as a Resource Manager

The operating system must:

- keep track of which program is using which resource,
- grant resource requests,
- delay or reject conflicting requests,
- protect users and programs from interfering with each other,
- account for resource usage,
- ensure fair and efficient sharing.

When multiple users are involved, the need becomes even greater because users may otherwise interfere not only with hardware but also with shared information such as files and databases.

Two Ways of Sharing Resources

Resource management often involves **multiplexing**, which means sharing a resource among many users or programs.

There are two main forms of multiplexing:

1. **Time Multiplexing**
2. **Space Multiplexing**

Time Multiplexing

Time Multiplexing means different programs or users take turns using the same resource over time.

Only one user or program uses the resource at a given moment, but many get a chance by using it one after another.

Example: CPU Sharing

If a system has only one CPU but many programs want to run, the operating system:

- gives the CPU to one program for a short time,
- then switches to another program,
- then to another,
- and later may return to the first one again.

The OS decides:

- who goes next,
- how long each one runs.

Example: Printer Sharing

A printer can also be time multiplexed. If many print jobs are waiting, the operating system decides which job will be printed next.

Time multiplexing means **taking turns**. It is commonly used for the CPU and printers.

Space Multiplexing

Space Multiplexing means dividing a resource into parts so that multiple programs or users can use it at the same time.

Instead of taking turns, each one gets a portion of the resource.

Example: Main Memory

Main memory is usually divided among several running programs. This allows multiple programs to remain in memory at the same time.

This is more efficient than giving all memory to a single program, especially when each program needs only a small part of it.

Example: Disk Space

Disk space is also space multiplexed. A single disk may hold files belonging to many users at once. The operating system must:

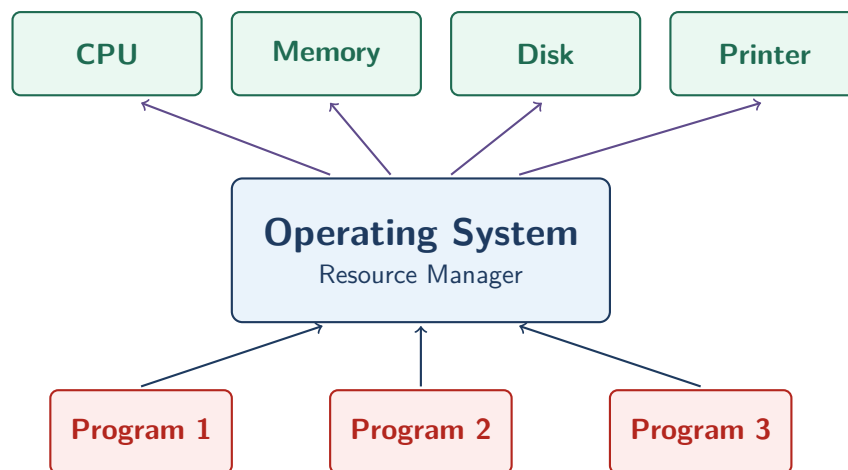
- allocate disk space,
- keep track of used blocks,
- know which user or file owns which part.

Space multiplexing raises issues such as fairness, protection, and tracking. These problems are handled by the operating system.

Time Multiplexing vs Space Multiplexing

Feature	Time Multiplexing	Space Multiplexing
Basic idea	Take turns using the resource	Divide the resource into parts
Usage style	One after another	At the same time
Examples	CPU, printer	Memory, disk
OS decision	Who uses next and for how long	Who gets how much space

Conceptual Diagram



The OS stands between programs and resources and manages their controlled allocation.